# Bala Harshith S

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**AI/ML Engineer | Production Machine Learning, GenAI and MLOps**

## PROFESSIONAL SUMMARY

- AI/ML engineer with 4+ years of experience building and deploying machine learning, NLP, recommendation, retrieval, and generative AI solutions across healthcare, financial services, and retail.
- Develops production LLM applications, retrieval-augmented generation pipelines, tool-using agents, semantic search, structured outputs, evaluation frameworks, and grounded enterprise workflows.
- Applies classification, regression, clustering, ranking, forecasting, anomaly detection, deep learning, feature engineering, experimentation, calibration, and explainability to real business problems.
- Builds scalable Python, SQL, Spark, API, data, training, inference, and monitoring systems using Azure, AWS, Databricks, SageMaker, Docker, Kubernetes, MLflow, and CI/CD.
- Owns delivery from problem framing and data preparation through modeling, testing, deployment, observability, iteration, and communication with product, engineering, and business stakeholders.

## TECHNICAL SKILLS

**Generative AI and Agents:** Large Language Models, RAG, LangChain, LangGraph, LlamaIndex, Prompt Engineering, Context Engineering, Tool Calling, Agent Orchestration, Structured Outputs, Embeddings, Vector Search, Hybrid Retrieval, Reranking, Grounding, Citations

**Machine Learning:** scikit-learn, PyTorch, TensorFlow, Hugging Face, XGBoost, Classification, Regression, Clustering, Ranking, Recommendation Systems, NLP, Deep Learning, Forecasting, Anomaly Detection, Feature Engineering, Model Evaluation

**Programming and Data:** Python, SQL, FastAPI, REST APIs, Spark, PySpark, Databricks, Airflow, pandas, NumPy, PostgreSQL, Snowflake, ETL/ELT, Batch Processing, Streaming, Git, Linux

**Cloud and MLOps:** Azure OpenAI, Azure AI Search, Azure ML, AWS, SageMaker, S3, Lambda, Docker, Kubernetes, MLflow, GitHub Actions, CI/CD, Model Serving, Experiment Tracking, Versioning, Monitoring, Drift Detection, Rollback

**Quality and Engineering:** A/B Testing, Cross-Validation, Calibration, Explainability, Error Analysis, Data Quality, Leakage Prevention, Unit Testing, Integration Testing, Observability, Logging, Tracing, Security, Privacy, Responsible AI, Technical Documentation

## PROFESSIONAL EXPERIENCE

### McKesson | AI/ML Engineer | Jan 2025 - Present

- Design and deploy production LLM, NLP, retrieval, and machine learning applications for healthcare workflows using Python, FastAPI, Azure OpenAI, Azure AI Search, LangChain, LangGraph, and Databricks.
- Build grounded RAG pipelines across ingestion, chunking, metadata enrichment, embeddings, hybrid retrieval, reranking, prompt and context assembly, structured outputs, citations, and permission-aware access.
- Improved retrieval accuracy and answer relevance by 35% through representative evaluation datasets, grounding checks, failure taxonomy, prompt experiments, retrieval tuning, and systematic error analysis.
- Develop tool-using agent workflows with planning, memory, reusable tool schemas, retries, fallbacks, human approval, trace logging, and bounded execution for reliable enterprise automation.
- Productionize containerized AI services with automated tests, CI/CD, versioning, monitoring, latency and cost analysis, regression gates, alerts, rollback procedures, and operational runbooks.
- Partner with product managers, engineers, data teams, security partners, and healthcare SMEs to define requirements, quality measures, risks, and measurable workflow outcomes from discovery through support.

### PNC Financial | Machine Learning Engineer | Dec 2022 - Nov 2023

- Built Python, SQL, Spark, and AWS machine learning systems across more than 12 million transaction and interaction records for risk prediction, anomaly detection, ranking, forecasting, and decision support.
- Developed and evaluated classification, regression, clustering, gradient-boosting, and neural-network models using scikit-learn, XGBoost, PyTorch, feature engineering, cross-validation, calibration, and explainability.
- Automated data preparation, training, evaluation, model packaging, deployment, and monitoring with SageMaker Pipelines, MLflow, S3, Lambda, Docker, and CI/CD, improving release consistency by 30%.
- Implemented leakage prevention, data-quality validation, feature consistency checks, subgroup analysis, drift indicators, threshold optimization, logging, alerts, regression testing, and rollback controls.
- Reduced distributed Spark processing from more than two hours to under 45 minutes through partitioning, caching, query optimization, and reusable transformations, accelerating experimentation and scoring workflows.
- Collaborated with risk, product, data, platform, and security teams to diagnose production issues, communicate model limitations and trade-offs, and connect technical metrics to operational decisions.

### Macy's | Data Scientist | Jun 2021 - Nov 2022

- Built recommendation, personalization, segmentation, forecasting, classification, and customer analytics solutions using Python, SQL, scikit-learn, XGBoost, collaborative filtering, and behavioral features.
- Developed PySpark, Databricks, pandas, NumPy, and SQL workflows for data cleansing, feature engineering, training datasets, batch scoring, experimentation, and customer-facing retail use cases.
- Created collaborative-filtering and segmentation models that converted sparse customer signals into personalized experiences and improved targeting accuracy by 25%.
- Designed offline evaluations and A/B test analyses using relevance, coverage, stability, engagement, conversion, segment performance, and business KPIs to guide iterative model improvements.
- Implemented reusable validation, scheduled jobs, monitoring, dashboards, and documentation that improved the reliability and reproducibility of analytical and model-driven workflows.
- Worked with product, marketing, merchandising, analytics, and engineering teams to frame problems, present model behavior and uncertainty, and translate findings into actionable product decisions.

## CERTIFICATIONS

- Oracle Cloud AI Infrastructure 2025

## EDUCATION

**Master of Science in Computer Science - University of Central Missouri**

**Bachelor of Technology in Electronics and Communication Engineering - Teegala Krishna Reddy Engineering College**